Game Analysis of Service Resource Pricing In Network Platform Environment

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***Abstract*-With the rise and popularity of the Internet, the transaction of service resources built on the platform keeps developing. The transaction process of service resources is the process of the game between the seller and the buyer on the quality and price. Based on the game theory, this study constructs a game model of service resource transaction, indicating that in the non-repetitive game, there are six possible game outcomes between seller and buyer, namely (high quality, high price, buy), (high quality, low price, buy), (high quality, high price, buy), (low quality, high price, buy), (low quality, low price, buy), (low quality, low price, buy). In the repeated trading game, there are only two possible game outcomes exist, namely (high quality, high price, purchase), (low quality, set low price, purchase). Finally, the model gives the pricing criteria of the seller's sales service resource transaction.**

Keywords-Service Resources, Signal Game, Transaction Price, Equilibrium

I. INTRODUCTION

In his book "Conflict of Security and Progress", the British economist Fisher divided the results of the industry into three stages: agriculture, industry, and service. Undoubtedly, China is currently in the third stage, that is, the rapid development of the service industry. The stage of rapid growth of the service economy. According to statistics from the Bureau of Statistics, the service industry accounts for 51.6% of GDP. The scope of service resources ranges from labor type to content type, data type and technology type. The service economy has developed to an unprecedented height. The service industry has become the largest industry in the national economy. The intangible, instantaneous, perishable, inseparable, complex and heterogeneous characteristics of service resources determine the difference between the pricing process and traditional commodities[1-3]. With the development of service economy, how to make reasonable and effective pricing of service resources becomes more and more important. All along, China has little research on the pricing of service resources, most of which focus on service marketing[4], value assessment[5-8], quality assessment[9], model innovation[10-11], from the perspective of gaming. There are very few documents to study its pricing. In reality, most of the service resources are based on the platforms of Zhu Bajie, 58 Home, and Yipin Witkey. The transaction is essentially the process of the buyer taking action based on the existing information after the seller's bid, and can be regarded as the participant 1 (Signal game between seller) and participant 2 (buyer). In summary, firstly, this paper uses signal game to build a game model of service resource pricing under the network platform environment. Secondly, it points out the possible game results exist between the buyer and the seller that in the repeated game and non-repeated game of service resource. Finally, it gives the pricing standard when the seller provides high or low quality service resource respectively.

II. BASIC HYPOTHESIS

In the actual situation, the participants of the game include the seller, the buyer, the platform and the government. The research focuses on the pricing of service resources. As far as the current market is concerned, the platform and the government have certain regulatory effects on the pricing of resources. However, the seller is still the decision maker of the price, and the buyer is the recipient of the price. Therefore, this article only uses the service resource transaction buyers and sellers as the game participants to construct the game model, and makes the following assumptions:

Hypothesis 1: It is assumed that the success of service resource transaction only depends on the buyer and seller participating in the game, and the market situation of service resource is not considered as an external factor.

Hypothesis 2: It is assumed that both buyers and sellers are rational people, that means both buyers and sellers will

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strive to maximize their own interests in this game, and then have the incentive to make rational decisions in the transaction to obtain potential benefits.

Hypothesis 3: At the same time, it is assumed that the quality information of the service resource belongs to the seller's private information, the buyer only knows the pricing of the service resource, and the buyer can only judge the quality of the service resource according to the pricing, that means the buyer has incomplete information about the quality of service resources sold by the seller.

Hypothesis 4: It is assumed that there are high and low quality of service resources in the market, the buyer's alternative strategies include purchase and non purchase. The utility of the buyer's choice to purchase high quality and low quality service resources is u_g and u_d respectively, and $u_g > u_d$. The utility obtained by the buyer's choice not to purchase is 0.

Hypothesis 5: It is assumed that the service resource seller's alternative strategy includes both high and low price. The high price is indicated by p_h, the low price is indicated by p_l, and $p_h > p_l$. The cost of the seller's efforts to have high and low quality service resources is divided into c_h and c_l. If the seller chooses to set a high price for low-quality service resources, the additional packaging promotion cost is c_p, and the penalty cost added by the market after being recognized by the buyer in the transaction is c_r.

Hypothesis 6: It is assumed that $u_g > p_h$, $u_d > p_l$ and $u_d < p_h$, otherwise according to the rational person hypothesis, the transaction would not have happened.

Hypothesis 7: When the quality of service resources is high and the seller chooses to set a high price, if the transaction is successful, the buyer's profit is $u_g - p_h$ and the seller's profit is $p_h - c_h$; if the transaction fails, the buyer's profit is 0 and the seller's profit is $-c_h$. When the quality of service resources is high and the seller chooses to set a low price, if the transaction is successful, the buyer's profit is $u_g - p_l$ and the seller's profit is $p_l - c_h$; If the transaction fails, the buyer's profit is 0 and the seller's profit is $-c_h$.

When the quality of service resources is low and the seller chooses to set a high price, if the transaction is successful, the buyer's profit is $u_d - p_h$ and the seller's profit is $p_h - c_l$; if the transaction fails, the buyer's profit is 0 and the seller's profit is 0. When the quality of service resources is low and the seller chooses to set a low price, if the transaction is successful, the buyer's profit is $u_d - p_l$ and the seller's profit is $p_l - c_l$; if the transaction fails, the buyer's profit is 0 and the seller's profit is $-c_l$.

Hypothesis 8: The buyer purchases high quality at high price, high quality at low price and low quality service resources at low price respectively, it's profit is greater than 0, taht is $u_g - p_h > 0$, $u_g - p_l > 0$, $u_d - p_l > 0$. When the buyer purchases low-quality service resources at a high price, the profit may be positive or negative, which means the value of $u_g - p_l$ may be greater than zero or less than zero.

In summary, the buyer's expected profit on purchasing is

$$E(u) = \mathrm{p}(g/h)(u_g - p_h) + \mathrm{p}(g/l)(u_g - p_l) + \mathrm{p}(d/h)(u_d - p_h) + \mathrm{p}(d/l)(u_d - p_l)$$

. $p(g/h)$ indicates the probability that the buyer will set a high price for high quality service; $p(g/l)$ indicates the probability of the buyer setting a low price for high quality service; $p(d/h)$ indicates the probability of the buyer setting a high price for low quality service; $p(d/l)$ indicates the probability of the buyer setting a low price for low quality service. Obviously, when $E(u) \geq 0$, the buyer chooses to buy; when $E(u) \leqq 0$, the buyer chooses not to buy.

III. GAME ANALYSIS

Considering that the service resource transaction is a dynamic process of information asymmetry and mutual decision making, the following service resource trading signal game model can be established:

(1) In the first stage, naturally N choose the product type. In this study, we assumed that the quality of service resources only exists in high and low cases. The product type set is represented by θ and $\theta = \{g, d\}$, where g indicates the high quality of service resources and d Indicates the low quality of service resources.Assume that the prior probability of high-quality service resources is represented by $\mathrm{p}(g)$, the prior probability of low-quality service resources is represented by $\mathrm{p}(d)$,and $0 \leq \mathrm{p}(g),\ \mathrm{p}(d) \leq 1$;

(2) In the second stage, the seller makes a pricing decision $p = \{p_h, p_l\}$ on the service resources according to the selection of natural N. Which p is the signal that the

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buyer can receive, p_h indicates that the seller has set a high price for the service resource, p_l indicates that the seller has set a low price for the service resource, and $p_h > p_l$.

(3) In the third stage, the seller does not know the natural choice N, that means it does not know the type of product θ, but knows the probability distribution $\mathrm{p}(\theta)$ of the product belongs to θ. When the buyer receives the signal p from the seller, the seller applies Bayesian development according to the signal to correct the prior probability, get the posterior probability of high-quality service resources and low-quality service resources $\mathrm{p}(\theta / p)$, then choose an action from the action set $A = \{a_1, a_2\}$ according to $\mathrm{p}(\theta / p)$, decide whether to buy or not, where a_1 means purchase, a_2 means not to purchase.

The above game process can be expressed as the following game tree:

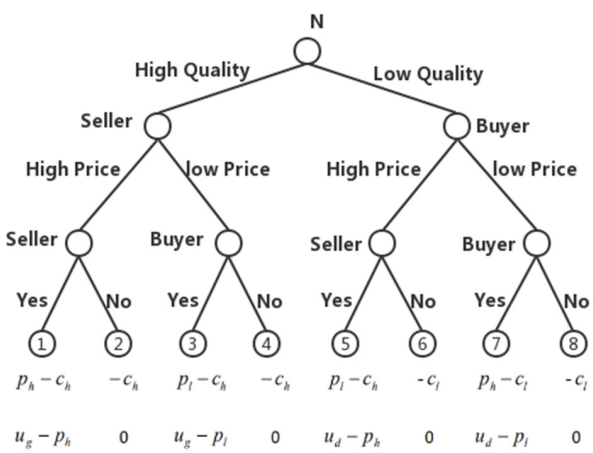

Figure 1 Service resource trading game model

In the figure, the first line in parentheses below ①- ⑧ indicates the seller's income, and the second line indicates the buyer's income.When the quality of the service resources is high, the s ,eller can choose to set the high price or the low price. When the quality of the service resources is low, the high price and the low price can also be selected.Therefore, the buyer cannot judge the quality of the service resources based solely on the pricing of the service resources. Therefore, the buyer must use the Bayesian rule in conjunction with the previous experience to make a correct judgment on the quality of the service resources.

IV. GAME ANALYSIS

A. Analysis of Participant 2 (Buyer):

(1) when the seller pays a high price, the buyer has two choices: to buy or not to buy. However, the buyer will eventually choose the decision-making method of maximizing the profit. Therefore, the final choice of the buyer is $Max\{(\mathrm{p}(g/h)(u_g - p_h) + \mathrm{p}(d/h)(u_d - p_h)), 0\}$.

1) When the buyer chooses to buy, there must be $\mathrm{p}(g/h)(u_g - p_h) + \mathrm{p}(d/h)(u_d - p_h) > 0$; From the hypothesis $(u_g - p_h) > 0; p(d/h) > 0$, we can get $\mathrm{p}(g/h)/\mathrm{p}(d/h) > (p_h - u_d)/(u_g - p_h)$.

2) When the buyer chooses not to buy, there is $\mathrm{p}(g/h)(u_g - p_h) + \mathrm{p}(d/h)(u_d - p_h) < 0$; And we can solve for $\mathrm{p}(g/h)/\mathrm{p}(d/h) < (p_h - u_d)/(u_g - p_h)$ by the same logic.

(2) When the seller sets a low price, the buyer also has two options: a_1 purchase and a_2 no purchase, and the choice is $Max\{(\mathrm{p}(g/l)(u_g - p_l) + \mathrm{p}(d/l)(u_d - p_l)), 0\}$.

1) When buyer chooses to buy, there must be $\mathrm{p}(g/l)(u_g - p_l) + \mathrm{p}(d/l)(u_d - p_l) > 0$; According to the assumption that $(u_g - p_l) > 0$, $(u_d - p_l) > 0$, $0 \le p(g/l)$ and $p(d/l) \le 1$, we can obtain $\mathrm{p}(g/l)(u_g - p_l) + \mathrm{p}(d/l)(u_d - p_l)$ is always greater than 0, when the seller chooses to set a low price, the buyer chooses to buy.

B. Analysis of Participant 1 (Seller):

Suppose the seller sets a high price, the probability that the buyer chooses to purchase is: $\mathrm{p}(\mathrm{a}_1/\mathrm{h}) = \mathrm{p}(g/h)/\mathrm{p}(d/h) > (p_h - u_d)/(u_g - p_h)$; the probability that buyer does not chooses to purchase is: $\mathrm{p}(\mathrm{a}_2/\mathrm{h}) = \mathrm{p}(g/h)/\mathrm{p}(d/h) < (p_h - u_d)/(u_g - p_h)$. When the seller sets the low price, the probability that the buyer chooses to buy: $\mathrm{p}(\mathrm{a}_1/l) = 1$; the probability that the buyer chooses not to buy : $\mathrm{p}(\mathrm{a}_2/l) = 1 - \mathrm{p}(\mathrm{a}_1/l) = 0$.

(1) When the quality of service resources is high, the seller has two kinds of decision-making methods, taht high price p_h and low price p_l, but the final choice is necessarily the biggest one among the sellers, so the final choice is $Max\{\mathrm{p}(a_1/h)(p_h - c_h) - \mathrm{p}(a_2/h)c_h,$

$\mathrm{p}(a_1/l)(p_l - c_h) - \mathrm{p}(a_2/l)c_h\}$, After simplified by $\mathrm{p}(a_1/l) = 1$, $\mathrm{p}(a_2/l) = 0$ and $\mathrm{p}(a_2/h) = 1 - \mathrm{p}(a_1/h)$, we get $Max\{\mathrm{p}(a_1/h)p_h - c_h,\ p_l - c_h\}$.

Therefore, in the case of high quality of service resources, if $\mathrm{p}(a_1/h)p_h - c_h > p_l - c_h$, then the seller sets a high price; if $\mathrm{p}(a_1/h)p_h - c_h < p_l - c_h$, the seller will set a low price.

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(2) When the quality of service resources is low, the seller also has a high price p_h and low price p_l two kinds of two decision-making methods, and thefinal choice is: $Max\{\mathrm{p}(a_1/h)(p_h - c_l - c_p) - \mathrm{p}(a_2/h)(c_l + c_p + c_r)$, $\mathrm{p}(a_1/l)(p_l - c_l) - \mathrm{p}(a_2/l)c_l\}$; After simplified by $\mathrm{p}(a_1/l) = 1$, $\mathrm{p}(a_2/l) = 0$, $\mathrm{p}(a_2/h) = 1 - \mathrm{p}(a_1/h)$, we get $Max\{\mathrm{p}(a_1/h)p_h - c_h,\ p_l - c_h\}$.

Therefore, in the case of low quality of service resources, if $\mathrm{p}(a_1/h) > (p_l + c_p + c_r)/(p_h + c_r)$, then the seller sets a high price. If $\mathrm{p}(a_1/h) < (p_l + c_p + c_r)/(p_h + c_r)$, the seller sets a low price.

V. SOLUTION OF MODEL

A. The solution of the buyer's game model

When the seller sets a high price, the buyer chooses to purchase with the probability of $\mathrm{p}(\mathrm{a}_1/\mathrm{h}) = \mathrm{p}(g/h)/\mathrm{p}(d/h) > (p_h - u_d)/(u_g - p_h)$, and does't chooses to purchase with the probability of $\mathrm{p}(\mathrm{a}_2/\mathrm{h}) = \mathrm{p}(g/h)/\mathrm{p}(d/h) < (p_h - u_d)/(u_g - p_h)$, when the seller sets a low price, the seller chooses to purchase.

B. Seller game model

On the premise of high quality of service resources, when the buyer decides that the probability of high price purchase is presented as $\mathrm{p}(a_1/h) > p_l/p_h$, the seller sets the high price; when the buyer decides that the probability of high price purchase is presented as $\mathrm{p}(a_1/h) < p_l/p_h$, the seller sets the low price(special case). On the premise of low quality of service resources, when the buyer decides that the probability of high price purchase is presented as $\mathrm{p}(a_1/h) > (p_l + c_p + c_r)/(p_h + c_r)$, the seller sets a high price; and the buyer decides that the probability of high price purchase is presented as $\mathrm{p}(a_1/h) < (p_l + c_p + c_r)/(p_h + c_r)$, the buyer sets a low price.

C. Repeated trading game analysis

It can be seen from the above analysis that when the product transaction is only performed once, whether it is a high-quality service resource or a low-quality service resource, the seller has the possibility of trading at a high price and a low price. However, in real market transactions, what exists between the seller and the buyer is often not a one-time transaction marketing, but a relationship marketing of repeated transactions.Therefore, when the buyer buys the high-priced products, he will feedback the information of the high-priced service resources to the market (whether there is any pricing fraud of low quality and high price). In this way, subsequent buyers continuously lower the probability of buying at high prices, that is $\mathrm{p}(a_1/h)$ keeps decreasing until the condition of $\mathrm{p}(a_1/h) > p_l/p_h$ cannot be met, and ultimately reduce the seller's behavior of setting high prices for service resources. When the quality of the service resources low, both sides information asymmetry makes the phenomenon of the "bad money drive out good money" (i.e., the seller to price low quality service resources cannot successful trading), meanwhile the seller can only through the lower the price to sales, so the entire service resource deals will step into a virtuous cycle track.

VI. SUMMARY

Through the game analysis of service resource transaction pricing, it is pointed out that there are five possible game outcomes in the independent one-time service resource trading game, namely: (1) high quality, high price, purchase; (2) high quality , set low prices, buy; (3) high quality, set high prices, do not buy; (4) low quality, set high prices, purchase; (5) low quality, set low prices, purchase. When the quality of service resources is high, the criterion for the seller to choose to set a high price or a low price is whether the probability of the buyer to buy at a high price $\mathrm{p}(a_1/h)$ is greater than the ratio of the low price set by the seller to the high price p_l/p_h ; when the quality of the service resource is low, the seller chooses to set a high price or a low price based on whether the probability of the buyer buying at a high price $\mathrm{p}(a_1/h)$ is greater than $(p_l + c_p + c_r)/(p_h + c_r)$. In the service resource market of repeated transactions (the market is valid), there are only two kinds of game results between the buyer and the seller, namely: (1) high quality, high price, purchase; (2) Low quality, fixed price, purchase.

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